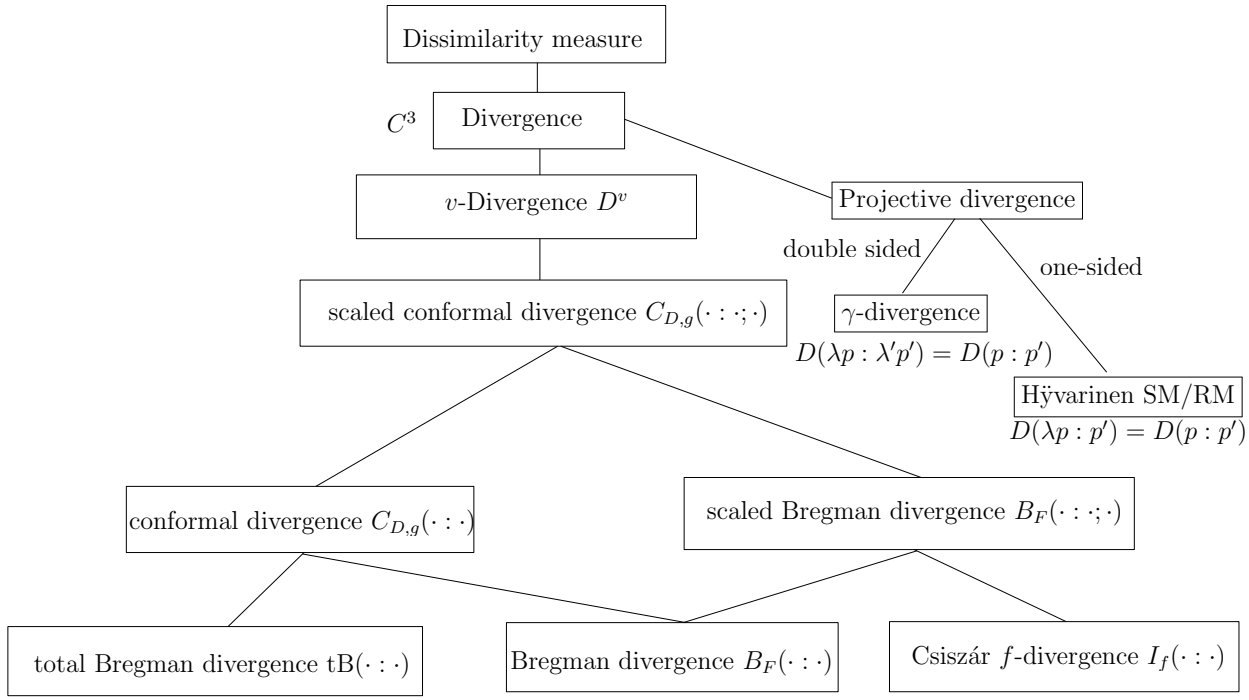

$D^v(P:Q) = D(v(P):v(Q))$

$I_f(P:Q) = \int p(x) f\left(\left(\frac{q(x)}{p(x)}\right)\right) \mathrm{d}\nu(x)$

$B_F(P:Q) = F(P) - F(Q) - \langle P-Q, \nabla F(Q)\rangle$

$\mathrm{tB}_F(P:Q) = \frac{B_F(P:Q)}{\sqrt{1+\|\nabla F(Q)\|^2}}$

$C_{D,g}(P:Q) = g(Q)D(P:Q)$

$B_{F,g}(P:Q;W) = W B_F\left(\frac{P}{Q}:\frac{Q}{W}\right)$

Figure 16: Principled classes of distances/divergences

post-conference edited books [75, 74]. We also mention the edited book [13] on the Occasion of Shun-ichi Amari's 80th birthday.

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A Monte Carlo estimations of f-divergences

Let (X, F, μ) be a probability space [60] with X denoting the sample space, F the σ-algebra, and μ a reference positive measure. The f-divergence [33, 90] between two probability measures P and Q both absolutely continuous with respect to a positive measure μ for a convex generator $f : (0, \infty) \to \mathbb{R}$ strictly convex at 1 and satisfying $f(1) = 0$ is

$$I_f(P : Q) = I_f(p : q) = \int p(x) f\left(\frac{q(x)}{p(x)}\right) \mathrm{d}\mu(x), \tag{205}$$

where $P = p\mathrm{d}\mu$ and $Q = q\mathrm{d}\mu$ (i.e., p and q denote the Radon-Nikodym derivatives with respect to μ). We use the following conventions:

$$0f\left(\frac{0}{0}\right) = 0, \quad f(0) = \lim_{u \to 0^+} f(u), \quad \forall a > 0, 0f\left(\frac{a}{0}\right) = \lim_{u \to 0^+} uf\left(\frac{a}{u}\right) = a \lim_{u \to \infty} \frac{f(u)}{u}. \tag{206}$$

When $f(u) = -\log u$, we retrieve the Kullback-Leibler divergence (KLD):

$$D_{\mathrm{KL}}(p : q) = \int p(x) \log \frac{p(x)}{q(x)} \mathrm{d}\mu(x). \tag{207}$$

The KLD is usually difficult to calculate in closed-form, say, for example, between statistical mixture models [96]. A common technique is to estimate the KLD using Monte Carlo sampling using a proposal distribution r:

$$\widehat{\mathrm{KL}}_n(p : q) = \frac{1}{n} \sum_{i=1}^{n} \frac{p(x_i)}{r(x_i)} \log \frac{p(x_i)}{q(x_i)}, \tag{208}$$

where $x_1, \ldots, x_n \sim_{\mathrm{iid}} r$. When r is chosen as p, the KLD can be estimated as:

$$\widehat{\mathrm{KL}}_n(p : q) = \frac{1}{n} \sum_{i=1}^{n} \log \frac{p(x_i)}{q(x_i)}. \tag{209}$$

Monte Carlo estimators are consistent under mild conditions: $\lim_{n \to \infty} \widehat{\mathrm{KL}}_n(p : q) = \mathrm{KL}(p : q)$.

In practice, one problem when implementing Eq. 209, is that we may end up potentially with $\widehat{\mathrm{KL}}_n(p : q) < 0$. This may have disastrous consequences as algorithms implemented by programs consider non-negative divergences to execute a correct workflow. The potential negative value problem of Eq. 209 comes from the fact that $\sum_i p(x_i) \neq 1$ and $\sum_i q(x_i) \neq 1$. One way to circumvent this problem is to consider the *extended f-divergences*:

Definition 7 (Extended f-divergence)**.** *The extended f-divergence for a convex generator f, strictly convex at 1 and satisfying $f(1) = 0$ is defined by*

$$I_f^e(p : q) = \int p(x) \left(f\left(\frac{q(x)}{p(x)}\right) - f'(1) \left(\frac{q(x)}{p(x)} - 1\right) \right) \mathrm{d}\mu(x). \tag{210}$$

Indeed, for a strictly convex generator f, let us consider the *scalar Bregman divergence* [23]:

$$B_f(a:b) = f(a) - f(b) - (a-b)f'(b) \geq 0. \tag{211}$$

Setting $a = \frac{q(x)}{p(x)}$ and $b = 1$ in Eq. 211, and using the fact that $f(1) = 0$, we get

$$f\left(\frac{q(x)}{p(x)}\right) - \left(\frac{q(x)}{p(x)} - 1\right) f'(1) \geq 0. \tag{212}$$

Therefore we define the *extended f-divergences* as

$$I_f^e(p:q) = \int p(x) B_f\left(\frac{q(x)}{p(x)} : 1\right) \mathrm{d}\mu(x) \geq 0. \tag{213}$$

That is, the formula for the extended f-divergences is

$$I_f^e(p:q) = \int p(x)\left(f\left(\frac{q(x)}{p(x)}\right) - f'(1)\left(\frac{q(x)}{p(x)} - 1\right)\right) \mathrm{d}\mu(x) \geq 0. \tag{214}$$

Then we estimate the extended f-divergence using importance sampling of the integral with respect to distribution r, using n variates $x_1, \ldots, x_n \sim_{\mathrm{iid}} p$ as:

$$\hat{I}_{f,n}(p:q) = \frac{1}{n}\sum_{i=1}^{n} f\left(\frac{q(x_i)}{p(x_i)}\right) - f'(1)\left(\frac{q(x_i)}{p(x_i)} - 1\right) \geq 0. \tag{215}$$

For example, for the KLD, we obtain the following Monte Carlo estimator:

$$\widehat{\mathrm{KL}}_n(p:q) = \frac{1}{n}\sum_{i=1}^{n}\left(\log\frac{p(x_i)}{q(x_i)} + \frac{q(x_i)}{p(x_i)} - 1\right) \geq 0, \tag{216}$$

since the extended KLD is

$$D_{\mathrm{KL}^e}(p:q) = \int \left(p(x)\log\frac{p(x)}{q(x)} + q(x) - p(x)\right) \mathrm{d}\mu(x). \tag{217}$$

Eq. 216 can be interpreted as a sum of scalar Itakura-Saito divergences since the Itakura-Saito divergence is scale-invariant: $\widehat{\mathrm{KL}}_n(p:q) = \frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{IS}}(p(x_i):q(x_i))$ with the scalar Itakura-Saito divergence

$$D_{\mathrm{IS}}(a:b) = D_{\mathrm{IS}}\left(\frac{a}{b} : 1\right) = \frac{a}{b} - \log\frac{a}{b} - 1 \geq 0, \tag{218}$$

a Bregman divergence obtained for the generator $f(u) = -\log u$.

Notice that the extended f-divergence is a f-divergence for the generator

$$f_e(u) = f(u) - f'(1)(u-1). \tag{219}$$

We check that the generator f_e satisfies both $f(1) = 0$ and $f'(1) = 0$, and we have $I_f^e(p:q) = I_{f_e}(p:q)$. Thus $D_{\mathrm{KL}^e}(p:q) = I_{f^e_{\mathrm{KL}}}(p:q)$ with $f^e_{\mathrm{KL}}(u) = -\log u + u - 1$.

Let us remark that we only need to have the scalar function strictly convex at 1 to ensure that $B_f\left(\frac{a}{b} : 1\right) \geq 0$. Indeed, we may use the definition of Bregman divergences extended to strictly convex functions but not necessarily smooth functions [50, 126]:

$$B_f(x:y) = \max_{g(y)\in\partial f(y)} \{f(x) - f(y) - (x-y)g(y)\}, \tag{220}$$

where $\partial f(y)$ denotes the subderivative of f at y.

Furthermore, noticing that $I_{\lambda f}(p:q) = \lambda I_f(p:q)$ for $\lambda > 0$, we may enforce that $f''(1) = 1$, and obtain a *standard f-divergence* [8] which enjoys the property that $I_f(p_\theta(x) : p_{\theta+\mathrm{d}\theta}(x)) = \mathrm{d}\theta^\top I(\theta)\mathrm{d}\theta$, where $I(\theta)$ denotes the Fisher information matrix of the parameteric family $\{p_\theta\}_\theta$ of densities.